Cover Page

NILAI

நிலை · steadiness

Senses tremor, separates it from intended movement, suppresses it on demand – and trains steadiness back.

- **Problem Statement / Concept Title**
  NILAI – a non-invasive AI wearable for real-time hand-tremor detection, on-demand suppression and personalised steadiness training

- **Theme**
  Assistive Technology & Accessibility (also: Healthcare · Elderly Care)

- **Team Name (Registered on portal)**
  Nilai

- **Team ID (Registered on portal)**
  HSC|TN|00325

- **State/UT**
  Tamil Nadu · Sri Sairam Engineering College, Chennai



Real-time tremor detection | On-demand suppression | No surgery · No drugs

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Problem – Proposed Solution

Hack for Social Cause

CORE PROBLEM

Tremor takes away the steady hand – and the activity that helps

~100 M people worldwide live with essential tremor (1.3%) [1]

5.8 % of adults over 65 are affected – it rises with age [1]

~4.6 M estimated in India; eating, writing, holding become hard

The vicious loop
Tremor → stop exercise → weaker control → worse tremor [2][3]

Today's options: pills that help only some, or brain surgery (DBS) costing lakhs. Severity is judged once in a while with a pen-and-spiral test – nothing at home.

PROPOSED SOLUTION

NILAI is a washable forearm sleeve with a snap-in sensor pod. It reads muscle (EMG) and motion signals, separates tremor from intended movement with on-device AI, cancels it on demand with a gentle counter-pulse, and coaches steadiness through an app.

DETAILED EXPLANATION – HOW IT WORKS

Sense
3-ch dry EMG + 6-axis IMU + heart-rate on the forearm

Distinguish
On-device AI isolates the 4–12 Hz tremor from voluntary motion

Suppress
Timed counter-pulse (EMS) damps the tremor – only when the user asks

Train
Personalised exercises + daily tremor score for long-term control

HOW IT ADDRESSES THE PROBLEM

- Steady hands for eating, writing, dressing – on demand
- Makes exercise possible again → breaks the loop
- Daily objective tremor score instead of a rare spiral test
- Caregivers & doctors track progress remotely

INNOVATION & UNIQUENESS

- Reads the cause (muscle EMG), not just the visible shake
- Tremor-vs-intent AI cross-checks EMG with motion – ignores vehicle/tool vibration
- One closed loop: detect → suppress → train
- Non-invasive, affordable, built in India

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Technical Approach

TECHNOLOGIES TO BE USED

Hardware

ESP32-C6 (BLE + Wi-Fi 6) · MPU6050 6-axis IMU · 3-ch dry EMG · MAX30102 heart-rate · biphasic EMS driver · Li-Po

AI & signal processing

Python · NumPy/SciPy (band-pass, Welch PSD) · scikit-learn classifier · WFLC adaptive filter · MediaPipe/OpenCV · C++ firmware

App & backend

React Native app · Flask live dashboard · FastAPI + PostgreSQL · BLE streaming

Github link

github.com/<your-repo> ← add link

Demo video

<YouTube / Drive link> ← add link

Extent of usage of AI tools

Gen-AI assistants used only for code help, literature search & slide drafting. Signal pipeline, classifier and hardware designed, trained & tested by the team.

METHODOLOGY & PROCESS – DATA FLOW

Acquire — EMG + IMU + PPG on the forearm

Pre-process — Filter 3.5–12 Hz, PSD band-power, RMS

Classify — Tremor vs voluntary – EMG–gyro coherence

Act — EMS anti-phase pulse, user-triggered

Learn — App: tremor score + training plan

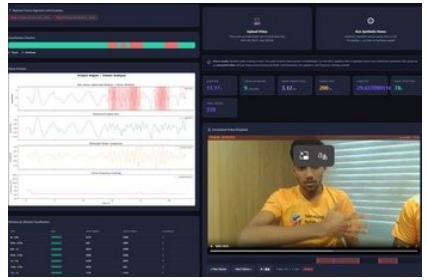
WORKING PROTOTYPE



Sensor pod: ESP32-C6, IMU, EMG, Li-Po



Live EMG waveform



Tremor-analysis dashboard

BUILD PROGRESS

STAGE 1 · DONE

Depth-camera + gyro tremor classifier

STAGE 2 · DONE

Signal-analysis dashboard (tremor vs intent)

STAGE 3 · NOW

Forearm EMG band with live waveform

STAGE 4 · NEXT

EMS suppression in the wearable pod + app

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Impact Potential

~100 M
people worldwide with essential tremor [1]

~4.6 M
estimated in India – most never see a specialist

1 in 17
adults over 65 affected – India is ageing fast [1]

Lakhs → ₹40K
DBS brain surgery vs. a NILAI device

Potential impact on the target audience

- People with essential tremor & Parkinson's, and the elderly
- Eat, write, dress and work independently again
- Return to exercise – which itself reduces tremor [3][4][5]
- Daily progress score builds confidence & adherence

Benefits of the solution

Social: dignity, independence, inclusion of elders; less caregiver burden

Economic: a fraction of surgery cost; shared kits for rehab centres & PHCs

Health system: remote monitoring cuts repeat clinic visits

Jobs: local assembly of sleeves & electrode kits

Stakeholders

- Patients, families & caregivers
- Neurologists & physiotherapists
- Rehab centres, PHCs & elder-care homes
- Govt. health schemes & NGOs
- CSR funders – kits for rural centres

Aligned with UN SDGs

SDG 3 · Good health & well-being | SDG 8 · Decent work | SDG 9 · Innovation | SDG 10 · Reduced inequalities

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Feasibility and Risk Analysis

Hack for Social Cause

ANALYSIS OF THE FEASIBILITY

Technical

Off-the-shelf parts; classifier, dashboard and EMG band already working (3 of 4 stages).

Economic

Target ₹40K device + annual service & app subscription; CSR funds rural kits.

Operational

Pilot with Chennai rehab & physiotherapy centres; app-based remote follow-up.

Validated

Concept recognised as a Samsung Solve for Tomorrow 2026 national finalist.

POTENTIAL CHALLENGES & RISKS → STRATEGIES FOR OVERCOMING THEM

Challenge / risk	Level	Strategy to overcome
False triggers from vehicle / tool vibration at tremor frequency	High	Act only when EMG (muscle origin) and gyro agree – coherence check
EMS safety & user comfort	High	Current-limited driver, clinician-set limits, user-triggered only, physical stop button
Noisy dry-electrode EMG	Medium	Compression sleeve, reference electrode, signal-quality gate before any action
Medical-device approval (CDSCO)	Medium	Launch as monitoring & training aid; gather pilot data before suppression claims
Affordability in rural India	Medium	Shared CSR-funded kits at PHCs & rehab centres; B2B service model
Health-data privacy	Low	On-device processing; consent-based sharing as per DPDP Act 2023

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Team

TEAM COMPOSITION & ROLES

YR

Yuvabharathi R

Team Lead

User research, clinical outreach, project management & pitch

Sri Sairam Engineering College

SH

S J Hema Priya

Hardware & sensing

EMG / EMS circuits, sensor integration, sleeve & pod design

Sri Sairam Engineering College

MK

Manoj Kumar K

AI/ML & embedded

Signal pipeline, tremor classifier, ESP32 firmware, dashboard & app

Sri Sairam Engineering College

CHALLENGES WE FACED

- **Raw EMG too noisy →** pivoted to depth-camera + gyro first, then a dry-electrode compression band
- **Vibration looks like tremor →** added EMG–gyro coherence so only muscle-driven tremor counts
- **Electronics vs. washable wear →** detachable pod with pogo-pin contacts and magnetic alignment

OUR JOURNEY



Idea	Prototype	Wearable	Recognition	Next
Tremor ↔ inactivity loop	Classifier + dashboard	EMG band, live signals	Samsung SFT 2026 finalist	EMS pod + clinical pilot

Building an affordable, made-in-India way for people with tremor to stay independent and active.

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